COMPUTER FUNDAMENTALS

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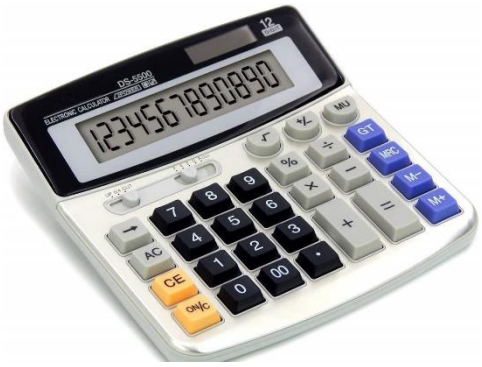
INTRODUCTION

Today, almost all of us in the world make use of computers in one way or the other. You will find its applications in various fields like *engineering, medicine, commercial, research and others*.
Not only in these sophisticated areas, but also in our daily lives, computers have become indispensable. They are present everywhere, in all the devices that we use daily like *cars, games, washing machines, microwaves etc*. and in day-to-day computations like *banking, reservations, electronic mails, internet* and many more.

The word **computer** is derived from the word **compute.** Compute means to calculate.



The computer was originally defined as ***a super-fast calculator***. It had the capacity to solve complex arithmetic and scientific problems at extremely high speed. But nowadays in addition to handling complex arithmetic computations, computers perform many other tasks like *accepting, sorting, selecting, moving, comparing* various types of information. They also perform arithmetic and logical operations on alphabetic, numeric and other types of information.

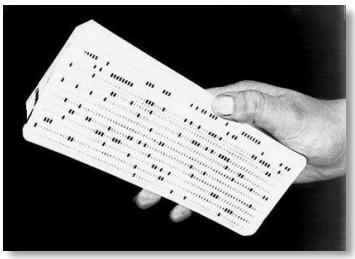
Computer Generations

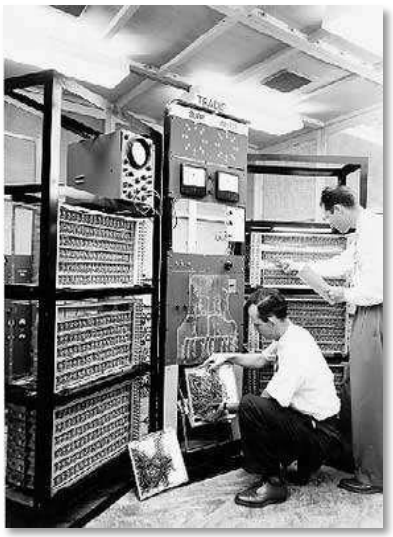
Generation in computer terminology ***is a change in technology a computer is/was being used***.

Initially, the generation term was used to distinguish between ***varying hardware technologies***. Nowadays, generation includes ***both hardware and software***, which together make up an entire computer system.

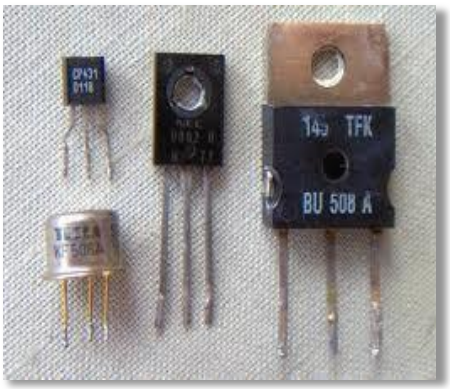
Following are the main five generations of the computer system:

- **First Generation (**1946-1959):
  - This generation relied on machine language to perform operations.
  - They could only solve one problem at a time.
  - Input was based on punched cards and paper tape.
  - Output was displayed on printouts.





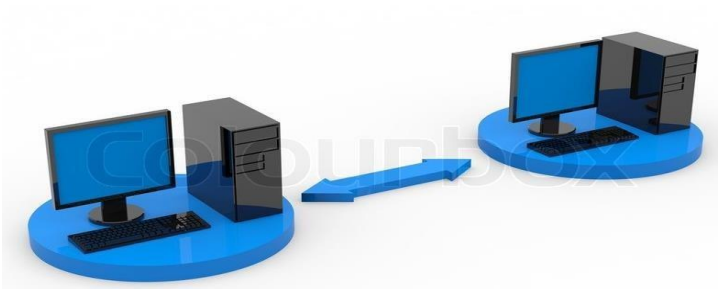
- **Second Generation (**1959-1965):
  - The transistor was far superior to the vacuum tube
  - Computers to become smaller, faster, cheaper, more energy-efficient and more reliable.
  - High-level programming languages.
  - It is the first computers that stored their instructions in their memory.



- **Third Generation (**1965-1971):
  - Integrated Circuit based.
  - Users interacted with the computers through keyboards and monitors.
  - Interfaced with an operating system.

❖ Fourth Generation (1971-1980):

- VLSI technology used
- Very cheap
- Very small size
- Use of PCs
- Concept of internet was introduced
- Great developments in the fields of networks
- Computers became easily available

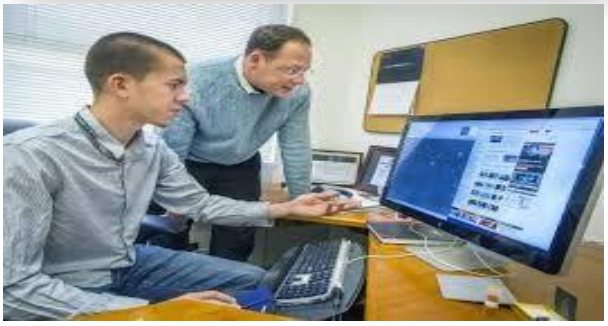


❖ Fifth Generation (1980-onwards):

- ULSI microprocessor based.
- Parallel Processing.
- Natural language processing
- More user-friendly interfaces with multimedia features.
- Very powerful and cheaper.

APPLICATIONS OF COMPUTERS

Today computers find widespread applications in all activities of the modern world. Some of the major application areas include:



1. **Military.**
2. **Scientific, Engineering and Research.**
3. **Business.**
4. **Information:** This is the age of information. Television, Satellite communication, Internet, networks are all based on computers.
5. **Education.**
6. **Games and Entertainment.**
7. **Communication.**
8. **Healthcare:** Computers have become an important part in hospitals, labs, and dispensaries. They are being used in hospitals to keep the record of patients and medicines. It is also used in scanning and diagnosing different diseases. ECG, EEG, ultrasounds, CT scans, etc. are also done by computerized machine.

Advantage of Computer

1. **Speed:** Smaller computers can execute thousands of instructions per second, while the more complex machines can execute millions of instructions per second.
2. **Accuracy:** Computers can execute hundreds of instructions without any errors.
3. **Efficiency:** The computers can perform repeated tasks with the same efficiency any number of times without exhausting their selves.
4. **Storage Capability:** Computers can store large amounts of data.
5. **Versatility:** They can perform non numerical operations fielding like *air- line reservation, electricity bills, data base management etc.*

Limitations of Computers:

Although the computers of today are highly intelligent and sophisticated, they have their own limitations.

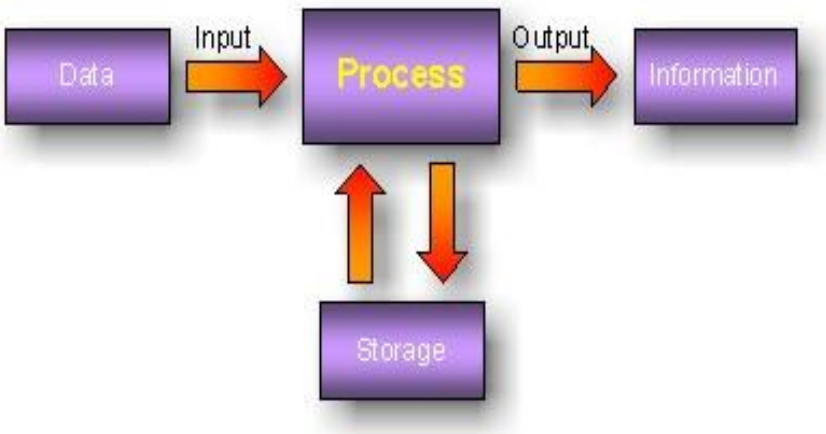
1. **No intelligence:** A computer cannot take any decision on its own.
2. **Dependency:** It functions as per the user's instruction thus it is fully dependent on humans.
3. **Environment:** The operating environment of the computer should be dust free and suitable.
4. **Feeling:** It cannot make judgment based on feeling, taste, experience, and knowledge unlike humans.

Computer Definition and Functions

It is a machine or device that performs processes, calculations and operations based on instructions provided by a software or hardware program.

<u>A computer has four functions:</u>

1) accepts data
2) processes data
3) produces output
4) stores results (information)



Types of computers by purpose

There are two types of computers according to their purpose:

1. **<u>Special-purpose Computers</u>** • Used for special purposes and usually expensive

- Heart monitoring equipment
- Traffic lights control system.
- Weather forecasting.
- Oil exploration.
- Automotive industries.
- Keeping time in a digital watch.
- Global positioning satellite (GPS) navigation tools
- Microwaves and other home appliances

2. **<u>General-purpose Computers</u>** • Most computers in use today are General-Purpose computers • It is more popular as they are cheap and can perform a variety of work. i.e. Personal Computer.

Computers Classification According to their size

1) **Supercomputer** Supercomputers are one of the fastest computers currently available.
Supercomputers are expensive and are employed for specialized applications that require huge number of mathematical calculations.
For example:

- Weather forecasting.
- Scientific simulations.
- Nuclear energy research.

2) **Mainframe**

Mainframe is large and is an expensive computer capable of supporting hundreds or even thousands of users simultaneously.

For example:

- Banks.
- Educational institutions.
- Control massive networks (Server).

rlg Institute of Technology

3) **Microcomputer (Personal Computer)**

The Micro-computers are specially designed for general purpose and to be used by one person at a time. Personal Computer is the smallest with low storage capacity and processing speed. It supports only one user at a time. Personal Computers are made up of input devices, output devices, storage device and one Processor.

Some Types of Microcomputers

3.1) Desktop (PC)

- Plays the major role in the user's life because users can perform several complicated tasks in few times without getting any difficulty.
- This computer is placed on the table, and it is connected to various components through wires such as keyboard, **mouse**, and **monitor** and system unit.
- These computers are capable to link internet through Wi-Fi or LAN cable.
- But it has some drawbacks such as bigger size compare to other types of Microcomputers.
- It is not portable because it needs fixed area.

3.2) Laptop

- It is designed like as stylish briefcase.
- They can also perform various complex functions like as Desktop Computer, and they able to run on their in-built battery as well as wall outlet supply.
- More expensive to desktop computers because in which various small size components are in built such as keyboard, touchpad LCD display, and other internal parts (Motherboard, CPU, Hard disk, and more).
- Its best advantage is portability.

3.3) Tablet microcomputers

- Handheld portable devices along with touch screen interface, and they are small size than laptop devices but bigger to Smartphone.
- Users can perform both activities (Input/output) on its LCD screen.
- On the Tablet, several applications such as word processing and spreadsheets can be run. Sometimes.
- Users can get problems due to lack of its keyboard, but they can use external keyboard.

3.4) Smartphone

- It is a touch screen mobile phone that is capable to perform various advance functions similar to computer such as install **operating system**, download different apps, internet access, and more activities.

What is a computer system?

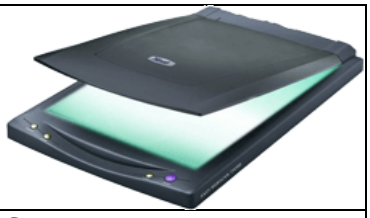
A computer system is made up of 4 main types of components:

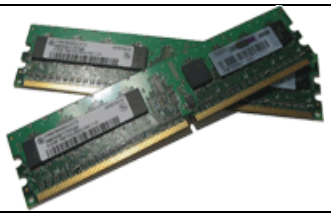
1) Hardware

Hardware is 'physical' means that it can be touched and picked up. Hardware works with computer code (Software) to carry out jobs such as:

- Producing an image on a computer screen
- Producing sound
- Moving cursers / pointers around a screen
- Etc.

Examples of hardware Devices:

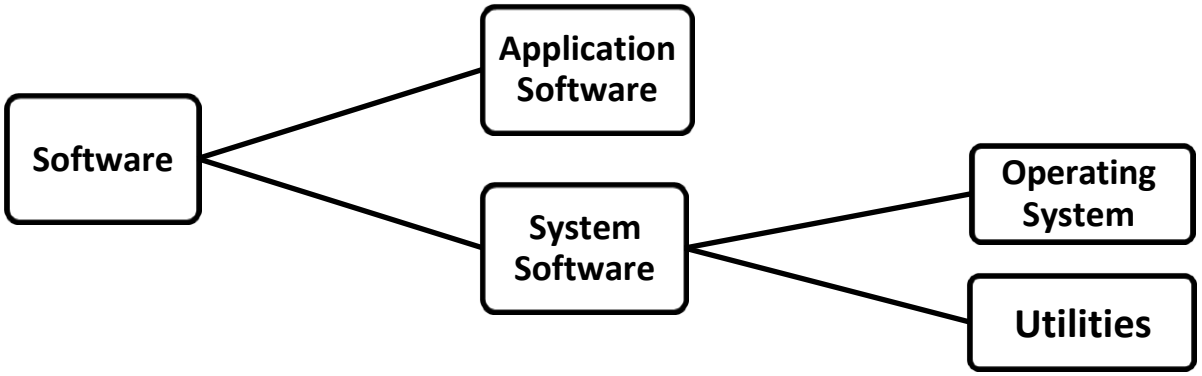
			
Keyboard	Mouse	Monitor	Scanner

Printer	CD	Motherboard	Digital camera
			
Speakers	Web came	Memory Card	Hard disk
Processer	Microphone	Flash memory	

2) Software (Software: Bringing the Machine to Life)

- Software cannot be touched or picked up, not physical.
- It exists as computer code inside the computer.
- Software tells the hardware what to do.
- Without software, most hardware would be useless.

Types of software:



2.1) Application software:

Application software tells the computer how to accomplish tasks the user requires.

Some important kinds of application software are:

- Word processing programs
- Spreadsheets
- presentation programs
- graphics programs
- Web browsers
- Internet applications.

2.2) System software:

System software often manages the computer's resources.

2.2.1) **Utility software** it is a small system software designed to help the computer to run mor effectively and efficiently.

For Example:

- Anti-virus utilities scan for computer viruses.
- Backup software can make copies of all information stored on a disk.

2.2.2) **Operating Systems:** are the most important programs that run on a computer. Every general-purpose computer must have an operating system to run other programs.

3) Data

- Computer data processed by the CPU is stored in files and folders on the computer's hard disk.
- These files may be in the form of ***(text documents, images, audio clips, software programs, or other types of data)***.
- The smallest unit of measurement used for measuring data is **a bit**. A single bit can have a value of either 0 or 1. It may contain a binary value (such as On/Off or True/False), but nothing more.

- File sizes are often measured in **kilobytes**. Larger files, such as images, videos, and audio files, contain millions of bytes and therefore are measured in **megabytes**.
- **A byte**, or eight bits, is used as the fundamental unit of measurement for data. A byte can store 28 or 256 different values, such as letters, numbers, and symbols.
- A Modern storage device is typically measured in **gigabytes** or even **terabytes**. Larger units of measurement are usually reserved for measuring the sum of multiple storage devices or the capacity of large data storage networks.

NOTE: A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes. This is an important distinction, since a byte is 8x as large as a bit.

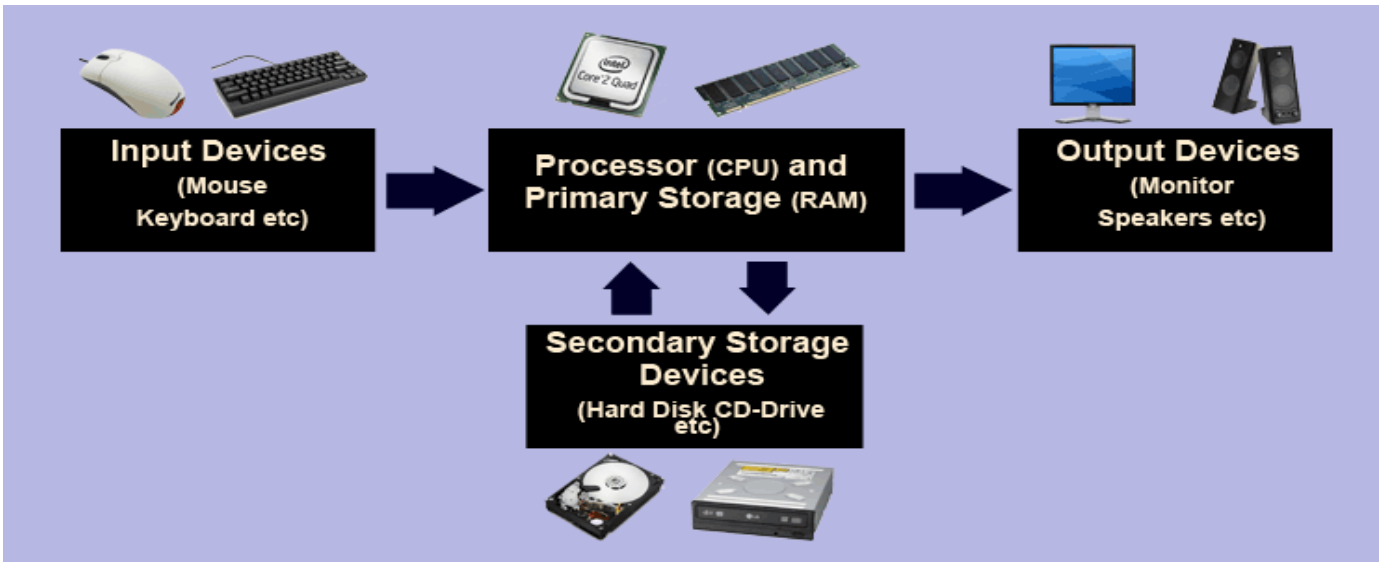
Unit	Value	Size
bit (b)	**0 or 1**	**1/8 of a byte**
byte (B)	**8 bits**	**1 byte**
kilobyte (KB)	**10001 bytes**	**1,000 bytes**
megabyte (MB)	**10002 bytes**	**1,000,000 bytes**
gigabyte (GB)	**10003 bytes**	**1,000,000,000 bytes**
terabyte (TB)	**10004 bytes**	**1,000,000,000,000 bytes**

4) Users

All the persons (users) that use the computer can be considered as live ware. EX: Programmers, Graphic Designer, Data entry operator, Hardware technician, Network engineer, Data base Administrator, Web Designer, System Analyzer

Hardware Component

1. Input Devices (keyboard, mouse etc)
2. Output Devices (monitor, speakers etc)
3. Secondary Storage Devices (hard disk drive, CD/DVD drive ,Flash Memory etc.)
4. Processor and Primary Storage Devices (CPU, RAM)



1) INPUT DEVICE

What is Input? Everything we tell the computer is Input.

Types of Input:

- Data (image from a Digital Camera, letters types via keyboard)
- Programs
- Commands
- User response

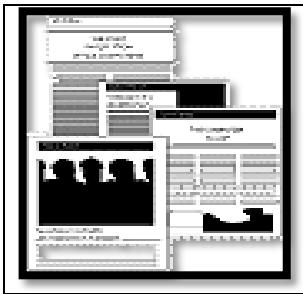
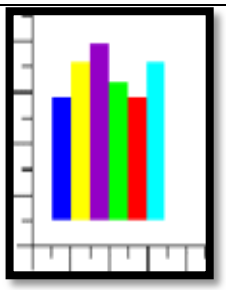
Some input devices are:

- Keyboard
- Mouse
- Microphone
- Bar code reader
- Graphics tablet

2) OUTPUT DEVICE

What is Output : it is data that has been processed into useful form.

Categories of Output:

		
Text documents	Graphics	Multimedia

Output device is used to send data out of the system. Following are few of the important output devices which are used in a computer.

Basic Output devices are:

2.1) Monitors

Monitors, commonly called as Visual Display Unit (VDU), are the main output device of a computer. It forms images from tiny dots, called pixels that are arranged in a rectangular form. The sharpness of the image depends upon the number of pixels.

There are two kinds of viewing screen used for monitors.

- Cathode-Ray Tube (CRT)
- Flat- Panel Display

Flat- Panel Display

Cathode-Ray Tube (CRT)

2.2) Printers

Printer is an output device, which is used to print information on paper.

Types of printers in use:

- **Laser Printers**

These are non-impact page printers. They use laser lights to produce the dots needed to form the characters to be printed on a page.

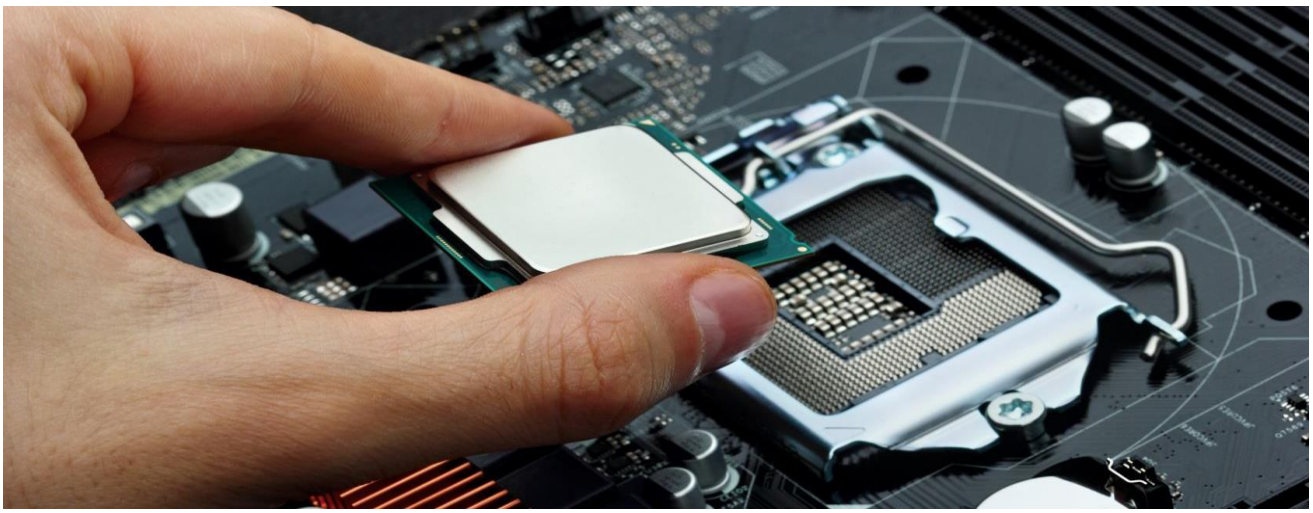
- **Inkjet Printers**

Printers based on a relatively new technology. They print characters by spraying small drops of ink onto paper. Inkjet printers produce high quality output with presentable features.

3) Processor and Primary Storage Devices

3.1) CPU Central Processing Unit

- The Central Processing Unit (CPU) is the brain of the computer.
- The CPU 'controls' what the computer does and is responsible for performing calculations and data processing.
- It handles the movement of data to and from system memory.
- CPU's come in a variety of speeds which are known as 'clock rates'. Clock rates are measured in 'Hertz'. Generally, the faster the clock rate, the faster the performance of the computer.



3.2) Main Memory

A memory is just like a human brain. Computer memory is the storage space in computer where data is to be processed and instructions required for processing are stored.

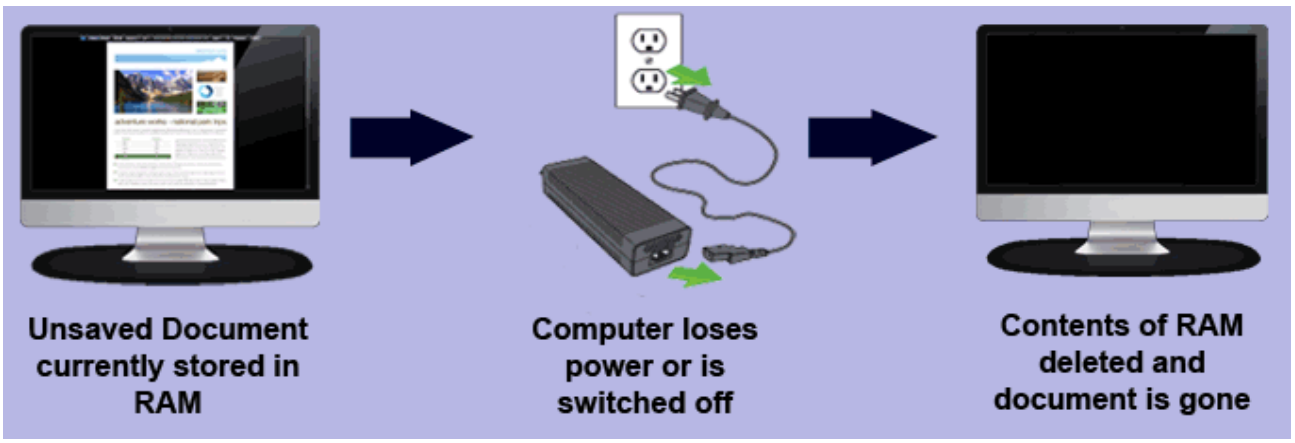
Primary Memory (MainMemory)

Primary memory holds only those data and instructions on which computer is currently working. It has limited capacity and data is lost when power is switched off. It is divided into two subcategories RAM and ROM.

3.2.1) RAM (Random Access Memory)

- RAM is used to temporarily store information that is currently in use by the computer. This can include anything from word documents to videos.
- RAM can be read from and written to and so the information stored in RAM can change all the time.
- RAM is a fast memory. Data can be written to and read from RAM very quickly.
- RAM is generally measured in GB (Gigabytes).
- RAM is Volatile Memory. This means that information stored in RAM is deleted as soon as the computer is turned off.
- The more RAM you have installed in your computer the faster it can perform.

***NOTE:* If you use up your entire RAM by opening too many programs at once you will notice that your computer becomes very slow.**



3.2.2) ROM (Read Only Memory)

- ROM is used to permanently store instructions that tell the computer how to boot (startup).
- It loads the operating system (e.g. Windows). These instructions are known as the BIOS (Basic input/output system) or the boot program.
- Information stored in ROM is known as READ ONLY. This means that the contents of ROM cannot be altered or added to by the user.
- Data stored in ROM can be accessed and read very quickly.
- ROM is Non-Volatile memory, its content not lost when the computer loses power.

4) Secondary Storage Devices (hard disk drive, CD/DVD drive , Flash Memory etc.)

- External memory.
- It is slower than main memory.
- These are used for storing data/Information permanently.
- CPU directly does not access these memories instead they are accessed via input-output routines. Contents of secondary memories are first transferred to main memory, and then CPU can access it.
- Secondary storage devices permanently store data and programs for as long as we need.
- These devices are also used to back-up data in case original copies are lost or damaged.
- There are two categories of storage devices:

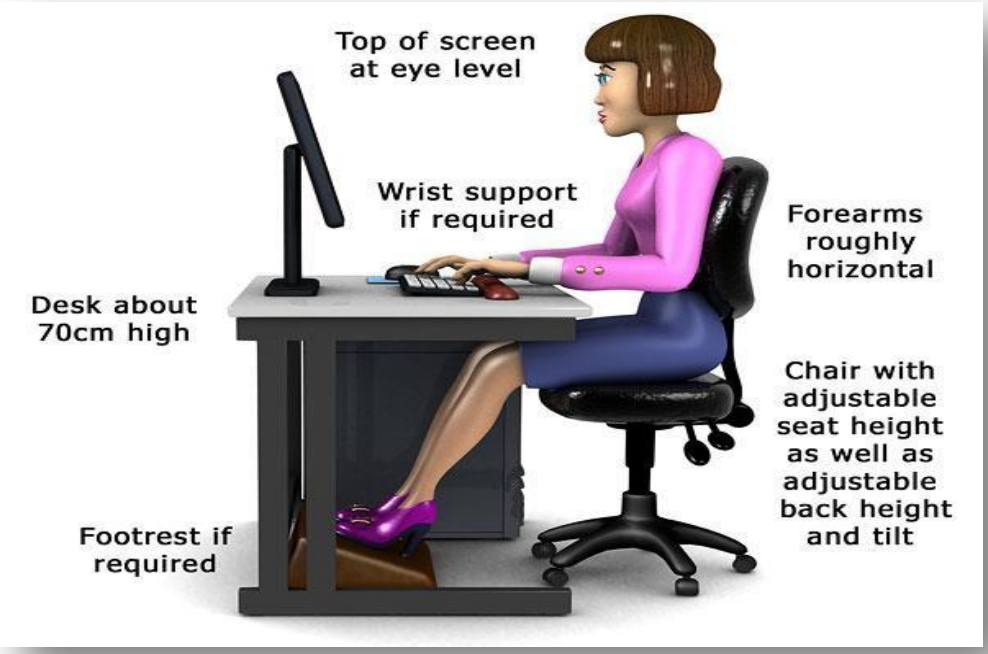
    - Internal Storage - Internal Hard Disk Drives
    - External Storage - External Hard Disk Drive, Memory Stick etc

External Hard Disk Drive

Internal Hard disk Drive

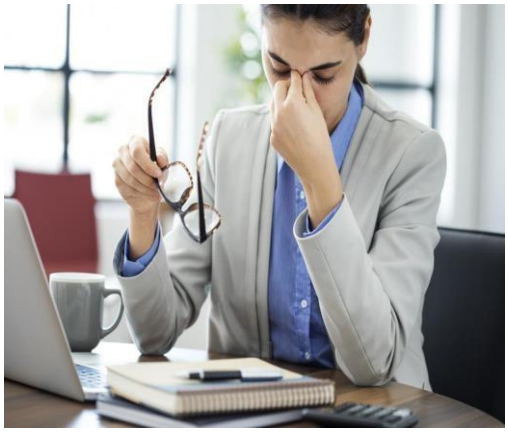
Worst Common Health Problems Caused by Computer Use

People nowadays are dependent on computers – research, communication, and productivity. However, ***scientists have found out that the use of computers affects a human's well-being***.



1. **Muscle Problems:** Regular computer users complain about muscle soreness and muscle fatigue. They also experience numbness in back, chest, arms, shoulders, and feet. The major cause of these is poor posture while using the computer.

2.* Vision Problems:** Too much focus on a computer screen is bad for the eyes. Prolonged exposure to computers, tablets, and smartphones cause ***computer vision syndrome. ***To avoid this condition, one should follow the 20-20-20 rule.***



2. **Headache:** Computer also caused headaches and migraines. ***There are various factors that trigger headaches, includes illumination, patterns, and images in the computer***. Also***, poor posture*** contributes to the onset of headaches.

3. **Stress:** Prolonged computer use also increased the risk of people to stress. If the person cannot tolerate stress, it will lead to depression and more serious mental health disorders.

4. **Obesity:** The use of computer makes people, especially children to be inactive, which will increase their risk to obesity and later on, to diabetes and heart disorders.

5. **Radiation:** Part of computers is radiation. Being exposed to radiation can increase tumors. It can also lead to miscarriage and birth defects. Most recently, scientists have found out that placing laptops in the lap affects male fertility. Computer radiation may not just decrease sperm count but will also trigger skin irritation.

Computer Ethics

Computer ethics is a system of ethical principles that a user must follow while using a computer at home, at school, or at work.

Guidance on how to abide by ethics when using a computer, as follows:

1. Use the computer in a manner that respects and considers others, and not to use it to harm them.
2. The user is keen not to overlap between his work and the work of others and not to delete files may needed by someone else.
3. Do not intrude on the work of another person on the computer.
4. Do not use the computer for stealing.
5. Do not copy or use software without paying for it and avoid the use of computer resources of others without permission.
6. Attention to the social consequences for the system or program writes by the computer user.
7. Avoid using the computer incorrectly may lead to ruin its parts, especially the computers in schools, universities, and public libraries.
8. Ensure the safety of the computer system by avoids using software's that overload or cause damage to the computer.
9. Maintaining the username and password and not to give it to others who are not authorized to use the devices.
10. Be careful not to use CDs or flash disks containing viruses and programs that lead to sabotage the computer.

Mobile Ethics

1. Close the mobile or make it in silent mode while attending meetings, seminars, conferences, in libraries, places of worship, aircraft, or waiting rooms.
2. Do not speak on the mobile aloud to not disturb others.
3. Avoid using an annoying and loud voice ringtone for mobile.
4. Do not listen to music or watch videos loudly and use speakers instead.
5. Do not discuss personal and work conversations in public places.
6. Choose the right time for others when communicating with them.
7. Do not send any offensive information that may cause inconvenience or harm to others in any way.
8. Do not take pictures or record an audio or video clip of others without their knowledge and without their consent, especially in places such as sports club, swimming pools, and when they are exposed to accidents and bad situations.
9. Do not access personal data in a person's mobile without his permission.
10. Do not see and read messages or receive a call while you are in a face-to-face conversation with someone.
11. Do not use mobile while eating a meal with someone.
12. Do not make calls or send text messages while driving.
13. Do not raise the sound of music too much when using earphones in public places, this may cause problems or accidents because of a person's attention to what is happening around him.

Respect copyright

1. The use of copyrighted material is prohibited without the author's permission. Such acts are in violation of laws and can cause judicial matters and criminal charges.
2. Illegal file sharing is prohibited by the Digital Millennium Copyright Act of 1998, and the new Millennium Act establishes rights to print and publish digital materials and procedures that can be followed in the event of a breach of these terms.
3. Respecting the privacy of others the most important ethic when using a computer. You must not do any form of altering and damaging the files or changing the password of others.
4. Respect the integrity of others' computing systems and avoid the use of invasion or violation programs to change the computer software's.
5. You must respect others and avoid harassing anyone or using someone else's account.

Licensing Computer Software

We have a lot of companies that manufacture software, and the largest company in the world in the software industry is Microsoft.

The giant operating systems in this area "Windows operating system" produced by this company occupy most of the computers in the world, other companies: IBM (IBM), Oracle, and Adobe.

The programs provided by these companies are either free or not, and the non-free programs are usually strong and to ensure that they are not break by copying, selling for commercial distribution, and use without purchase. These companies resort to a license.

This license is often in the form of a serial number that usually comes with the program when purchasing.

Crack

Crack is the infringement of the company's rights in the software that it produces by ignoring the company's license, and using the software without buying, copying, and distributing it, as if it were paid.

Crack is done by many means, including:

- The use of a ***small program called (crack)***, or (Patch); where this program makes some modifications to the other program that is intended to be used without purchase.
- ***The Keygen*** is a small program based on the generation and creation of false serial numbers that can be used in the license of the program.
- The company's rights to the software are sometimes broken by ***replacing one file with another within the program.***

COMPUTER THREATS (Malware)

1. **Malware** is a combination of the words malicious and software and is used to describe any type of bad code we may find on a computer. Any software that gets installed on your machine that performs some unwanted or harmful task, normally for the benefit of some third party, qualifies as malware.

2. **Virus:** A destructive program specifically written to destroy data or halt operation on computer systems. A true virus can attach itself to a file already on your system "host file". It can reproduce itself and infect computer systems.

3. **Trojan horses:** **A** malicious program that is written specially to cause damage or disturb the normal operation of a computer. They usually have a name that makes them appear to be a useful program that tricks a user into starting them.

4. **Worms:** Usually found on internets, these files would gather information (recording passwords when they were typed in or allowing for unauthorized accesses). Another type of worm is a file that just keeps replicating itself over and over make the computer slow.

5. **Spyware:** is a type of malware that can be installed on computers and collects little bits of information at a time about users without their knowledge and can be difficult to detect.

6. **Adware:** its purpose is to display those annoying pop-up advertisements. The purpose of adware is to get you to click on the ad it is presenting so that some third party may get paid just for the click.

Antivirus software

It is Program used to prevent, detect, and remove malicious software.

Securing Your System

There are several measures which you can take to secure your system.

1. **Personal firewall.**
2. **System Software Updates.**
3. **Be aware** of how viruses spread:
   - Do not open attachments in Emails unless you are sure they are legitimate.
   - Potential virus files that may have multiple extensions such as filename .txt .exe. If the extension ends in .exe, .com, or .bat do not double click on it or run it unless you are *SURE* it is from a legitimate source.
4. **Avoid installing bad applications** some computer programs may come with spyware or adware. Keep in mind that adware programs may download and install other programs from the internet. A personal firewall is one defense against this happening because it will normally notify you when a program accesses the internet.
5. **Configure your system** so you will see all file extensions as described on the page called "Windows File View Settings".
6. **Use anti-virus software:** Be sure to run anti-virus software and download updates at least twice per week.
7. **Choose strong passwords:** Devices and accounts must be protected from hackers by setting passwords that are hard to guess.

THANK YOU

LECTURER MYSAA HUSAM